

Smart Library Ontology Project Report v2

Knowledge Engineering and Ontologies Course Project Report

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Executive Summary

This project develops a Smart Library Loan Ontology for modeling circulation knowledge in a semantically structured way. The ontology represents members, librarians, resources, publishers, branches, loans, reservations, and resource statuses in OWL/RDF, and it is extended in Phase 2 with provenance-aware acquisition concepts, semantic topics, and lightweight personalization structures. The main objective is to move from a small conceptual ontology toward a more operational knowledge graph artifact that can support querying, validation, and future recommendation scenarios. The project uses Protégé-style ontology engineering principles, RDF modeling, SPARQL querying, and SHACL validation. Phase 2 additionally introduces a data acquisition perspective by documenting public API sources, local seed tables, preprocessing logic, and ontology population mappings. The project is also informed by recent research on ontology-driven automation and personalized semantic systems. The final outcome is an updated ontology version, supporting documentation, an expanded query set, SHACL shapes, and a report that explains how semantic modeling, data acquisition, and future LLM or predictive extensions can work together in a smart-library context.

Description of the Project

The main objective of the project is to design a semantic model for a Smart Library Loan System and to show how that model can support a knowledge graph capable of querying, validation, and future intelligent services. The real-world problem addressed by the project is that library knowledge is often fragmented across catalog systems, spreadsheets, digital repositories, and operational records. Without a formal semantic structure, it is harder to answer questions such as who borrowed a resource, which branch stores a printed book, which topics characterize a resource, or which data source supplied a metadata record. The project therefore uses

ontology engineering and Semantic Web technologies to define the domain in OWL, represent data as RDF triples, query the graph through SPARQL, and validate important constraints through SHACL. The implementation is centered around a library ontology that models resources, people, transactions, provenance, and topical enrichment. Phase 2 broadens the project by adding data acquisition concepts, member-profile concepts, and recommendation-ready constructs. The expected outputs are an updated ontology, a query file, a validation file, documentation, and a versioned specification. The intended application context is a library information environment in which semantic interoperability, explainable retrieval, and future AI grounding are important. The core competency questions guiding the design are: Which resources are currently available? Which resources are currently on loan? Which member borrowed a specific resource? Which branch stores a printed book? Which topics are linked to each resource? Which data sources were used to populate a resource? Which metadata records belong to the current acquisition batch? Which resources match a member's topic interests? Which recommendation candidates are available for a member? How many resources belong to each source or status category?

Ontology Design

The ontology is divided into a TBox and an ABox. The TBox defines the conceptual schema of the domain. The original ontology included classes such as ``Member``, ``Librarian``, ``LibraryResource``, ``Book``, ``PrintedBook``, ``EBook``, ``JournalArticle``, ``Loan``, ``Reservation``, ``LibraryBranch``, ``Publisher``, and ``ResourceStatus``. In Phase 2, the TBox was expanded with ``DataSource``, ``AcquisitionBatch``, ``MetadataRecord``, ``Topic``, ``MemberProfile``, ``InteractionEvent``, and ``Recommendation``. These additions allow the project to model provenance, topic-level semantics, and future personalization. The ABox contains sample individuals such as ``Member_Aylin_Demir``, ``Librarian_Emre_Kaya``, ``Book_OntologyDesign``, ``EBook_KnowledgeGraphs``, ``Loan_001``, and ``Reservation_001``. Phase 2 also adds provenance and personalization individuals such as ``DataSource_OpenLibraryAPI``, ``MetadataRecord_B5001``, ``Profile_Aylin_Demir``, and ``Recommendation_001``.

The key object properties include circulation relations (``loanedTo``, ``loanedResource``, ``reservedBy``, ``reservedResource``, ``processedBy``), bibliographic relations (``writtenBy``, ``publishedBy``, ``locatedAt``, ``hasStatus``), and new Phase 2 relations (``hasTopic``, ``describedBy``, ``retrievedFromSource``, ``generatedInBatch``, ``hasProfile``, ``hasInterestTopic``, ``recordsMember``, ``interactedWithResource``, ``recommendedFor``,

`recommendedResource`). The key datatype properties include `identifier`, `name`, `title`, `isbn`, `publicationYear`, `startDate`, `dueDate`, `returnDate`, `email`, `branchAddress`, `accessUrl`, `sourceUrl`, `retrievedAt`, `confidenceScore`, `interactionType`, and `profileScore`.

Version 2 also improves ontology reuse. Instead of remaining fully isolated, it aligns selected concepts with

Dublin Core and Schema.org. For example, `LibraryResource` is aligned with `schema:CreativeWork`, `Book` with

`schema:Book`, and the local properties `name` and `isbn` are aligned with `schema:name` and `schema:isbn`.

This creates lightweight interoperability without overcomplicating the model.

Research Integration

The selected research study for integration is Alrehaili et al. (2021), *Ontology-Based Smart System to Automate Higher Education Activities*. This study was selected because it presents a full ontology-centered

workflow that starts from heterogeneous data sources and ends with semantic querying and reasoning. It is

especially relevant to the current project because Phase 2 requires a clear Data Acquisition section and a

stronger explanation of how ontology development connects to real data. The study contributes an architectural

idea rather than a library-specific vocabulary: specification, knowledge acquisition, conceptualization, ontology development, and SPARQL-based use. We adapted this pattern directly to the smart-library domain. In

our project, the specification phase is represented by the competency questions and scope boundaries; the

knowledge acquisition phase is represented by public APIs and local seed files; the conceptualization phase is

represented by the added provenance, topic, and profile modules; and the ontology-development and SPARQL

phases are represented by the updated OWL model, SHACL shapes, and query file. The study therefore shaped the

system architecture, the reasoning-oriented ontology design, and the data-processing explanation in the report.

Mandatory Paper Review

The mandatory paper, Fernandez et al. (2023), proposes a personalized digital-library architecture that combines ontology engineering with a GRU-RNN model optimized through deep training tree and black widow

optimization techniques. The core methodology is hybrid: ontology is used to structure digital-library concepts and interrelations, while a predictive neural layer is used to classify or predict user behavior patterns. The main contributions are the use of a personalized ontology for the digital-library environment,

the integration of symbolic knowledge representation with predictive learning, and the optimization of the

recurrent model for better performance. For the Smart Library project, the direct algorithmic layer is not

implemented in this phase, but the conceptual idea is highly relevant. We adapted it by introducing ``MemberProfile``, ``InteractionEvent``, and ``Recommendation`` classes so that the ontology can later host the outputs of a predictive personalization model. In other words, our current system is not yet running a GRU-RNN, but it now has a schema that could support such a behavioral-prediction extension in future work.

Data Acquisition

The data acquisition strategy combines public bibliographic APIs, local seed tables, and project documentation. The Open Library API is used as a public bibliographic source for printed-book metadata, while the Google Books API is used for digital-resource enrichment. A manually curated department seed spreadsheet is used to reconcile identifiers, resource status labels, branch information, and controlled examples. Conceptual project documents such as the specification draft also function as domain sources for vocabulary selection and competency-question refinement. The collection methodology follows a small-scale version of a semantic ETL pipeline. First, target entities are identified from the ontology scope. Second, descriptive metadata are gathered from public and local sources. Third, values are normalized into tabular seed files. Fourth, the normalized values are mapped to ontology classes and properties. Fifth, provenance is preserved through ``DataSource``, ``AcquisitionBatch``, and ``MetadataRecord``.

Preprocessing includes identifier normalization, status normalization, topic normalization, and source reconciliation. Resource states were reduced to controlled individuals (``Available``, ``OnLoan``, ``Reserved``), while free-text topical descriptions were normalized to ``Ontology Engineering``, ``Knowledge Graphs``, and ``Semantic Web``. The quality limitations are that the current dataset is intentionally small and manually curated, and the project is a proof-of-concept rather than a production ingestion system. Even so, the acquisition workflow is explicit and academically defensible.

Knowledge Graph Construction

The knowledge graph is formed by combining the ontology schema with instance-level data represented as RDF triples. Each triple follows the subject-predicate-object model. Examples include:

- ``Loan_001 loanedTo Member_Aylin_Demir``
- ``Book_OntologyDesign hasStatus OnLoan``
- ``EBook_KnowledgeGraphs hasTopic Topic_KnowledgeGraphs``
- ``MetadataRecord_B5001 retrievedFromSource DataSource_OpenLibraryAPI``
- ``Recommendation_001 recommendedFor Member_Aylin_Demir``

The graph combines circulation facts, bibliographic information, provenance metadata, and interaction signals

in one semantic structure. In a deployment setting, this dataset can be loaded into a triplestore such as GraphDB, where namespace declarations, reasoning settings, and query endpoints can be configured. In the current repository, the graph is serialized in Turtle and RDF/XML, while the query file and SHACL file provide the operational layer for querying and validation. The knowledge graph therefore goes beyond a conceptual ontology because it now contains explicit data-acquisition and personalization individuals.

SPARQL Queries

The updated SPARQL file contains twelve queries. These queries cover three groups. The first group contains basic retrieval questions such as which resources are available, which resources are on loan, who borrowed a resource, which branch stores a printed book, and which topics are linked to each resource. The second group focuses on provenance and data acquisition, including which sources populated a resource and which metadata records belong to the Phase 2 acquisition batch. The third group covers profile-based retrieval and aggregation analytics, such as which resources match a member's interest topics, which recommendation candidates are available, how many resources belong to each status, how many resources came from each source, and how many interaction events belong to each event type. These queries are directly related to the competency questions and demonstrate that the ontology supports both descriptive and analytics-oriented use.

Validation

Validation is documented through SHACL in ``shapes/smart-library.shacl.ttl``. The shapes check important constraints that the ontology design expects to hold. ``LoanShape`` enforces one member, one resource, one librarian, and one due date per loan. ``ReservationShape`` enforces one member and one resource per reservation. ``LibraryResourceShape`` ensures that a resource has a title, a status, and a publisher. ``MetadataRecordShape`` ensures that each metadata record describes exactly one resource, is linked to at least one source, and has a confidence score between 0 and 1. ``RecommendationShape`` ensures that each recommendation points to exactly one member and one resource and that the score remains within a valid decimal range. These shapes are useful because they make data quality expectations explicit and prepare the project for reproducible validation in later stages.

LLM Integration

The course template requires LLM integration, and Phase 2 addresses this at the design level. The most realistic integration for this project is knowledge-grounded question answering or natural language to SPARQL translation. The ontology and knowledge graph provide the grounding layer, while an LLM can be prompted to transform a user question such as "Which available resources are about knowledge graphs?" into a query over the graph. Hallucination reduction is achieved by constraining answers to graph-backed results. A second LLM-related use is ontology population assistance, where an LLM can help normalize extracted titles, infer likely topics, or draft mapping candidates before human verification. Although a full production pipeline is not implemented in this phase, the ontology structure and provenance model now support such future integration much better than in Version 1.

Evaluation, Discussion and Conclusion

Technically, the project is stronger in Phase 2 because it now demonstrates more than a simple circulation ontology. The updated version introduces provenance-aware data acquisition, topic-level enrichment, profile-aware retrieval, and validation shapes. This improves the completeness and practical relevance of the ontology. The use of lightweight vocabulary reuse through Dublin Core and Schema.org also improves interoperability. The query layer is more mature because it now includes provenance and aggregation questions in addition to simple lookups.

The project still has limitations. The instance data remain intentionally small, the acquisition pipeline is illustrative rather than automated at scale, and the personalization layer is structural rather than predictive. The LLM integration is currently a design pathway instead of a fully implemented end-user system. However, these limitations are appropriate for the scope of a course project and are clearly documented rather than hidden.

In conclusion, the Smart Library Loan Ontology has evolved from a basic academic ontology draft into a better engineered Phase 2 semantic project. Its key contributions are the extension of the ontology with provenance and personalization concepts, the explicit documentation of data acquisition, the update of the competency-question and query layer, and the preparation of the model for future reasoning, validation, and AI integration. The project also demonstrates the educational value of ontology engineering as a bridge between raw data, structured semantics, and explainable intelligent behavior.

References and Format

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<https://www.w3.org/TR/shacl/>

Schema.org. (2026). **Schema.org vocabulary**. <https://schema.org/>

The Dublin Core Metadata Initiative. (2026). **DCMI metadata terms**.

<https://www.dublincore.org/specifications/dublin-core/dcmi-terms/>

Appendix

Repository artifacts

- GitHub repository: ``https://github.com/kemkem042/SmartLibraryOntology``
- Widoco documentation: ``https://kemkem042.github.io/SmartLibraryOntology/widoco/``
- Documentation landing page: ``https://kemkem042.github.io/SmartLibraryOntology/``
- Ontology: ``ontology/smart-library.ttl``
- RDF/XML export: ``ontology/smart-library.owl``
- Queries: ``queries/competency-questions.rq``
- SHACL: ``shapes/smart-library.shacl.ttl``
- Acquisition seeds: ``data/phase2/*.csv``
- Specification v2: ``docs/specification-v2.md``
- Research integration: ``docs/research-integration-v2.md``
- Mandatory review: ``docs/mandatory-paper-review-v2.md``